
Adaptive Basis LoRA: Temporal-Complexity-Scaled PEFT for Video-LLMs

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Abstract

1 We address a central bottleneck in video-language models: temporal reasoning
2 under strict token and parameter budgets. Existing parameter-efficient fine-tuning
3 (PEFT) methods such as LoRA and DoRA enforce a fixed small rank across time,
4 compressing complex motion into a handful of shared directions and encouraging
5 aggressive frame dropping, which harms temporal consistency and boundary lo-
6 calization Hu et al. [2021], Liu et al. [2024]. We propose Adaptive Basis LoRA
7 (AB-LoRA), a plug-and-play adapter that replaces the single static low-rank update
8 with a video-adaptive mixture of bases. For each video, we estimate a variable num-
9 ber of orthogonal temporal bases in CLIP latent space using EM with MDL/BIC
10 selection and train a lightweight temporal encoder to predict per-time-step respon-
11 sibilities Li et al. [2023]. Each attention block mixes basis-specific directions and
12 magnitudes in a DoRA-style decomposition, with Householder-inspired compres-
13 sion for parameter reduction Liu et al. [2024], Dong et al. [2024]. Training proceeds
14 in three stages: feature alignment, basis pretraining with an MDL-GMM teacher on
15 WebVid-scale data Bain et al. [2021], and instruction tuning with the basis encoder
16 frozen. We provide a full experimental plan covering datasets, metrics, fairness
17 controls, robustness grids, and ablations, plus PyTorch skeletons. A minimal syn-
18 thetic sanity check verifies end-to-end feasibility; it is not a benchmark evaluation.
19 AB-LoRA establishes a principled route to temporally adaptive PEFT that can be
20 layered on existing Image-LLMs to enable consistent long-video understanding.

21 1 Introduction

22 Large multimodal language models have become the backbone of video-language systems, yet they
23 must reason over long and irregular temporal dynamics under tight context windows and memory
24 constraints. Contemporary pipelines often process fewer than 16 sparsely sampled frames to remain
25 tractable, which sacrifices temporal fidelity. Parameter-efficient fine-tuning (PEFT) is attractive
26 compared to full model tuning, but dominant variants such as LoRA and DoRA apply a fixed low rank
27 to all time steps within each adapted projection Hu et al. [2021], Liu et al. [2024]. Although DoRA
28 improves stability by decoupling magnitude and direction, neither adapts the available capacity to the
29 temporal complexity of the input. This mismatch forces dynamic, long videos to compress motion
30 variability into few shared directions, often manifesting as temporally inconsistent answers and poor
31 localization of event boundaries-issues echoed in recent video-language studies focused on temporal
32 understanding Yang et al. [2022], Yu et al. [2023], Huang et al. [2023].

33 We hypothesize that temporal reasoning under token and parameter constraints requires adaptation
34 capacity that scales with input complexity. We introduce Adaptive Basis LoRA (AB-LoRA), which
35 replaces a single static low-rank adapter with a per-video mixture of bases. For each video, we obtain
36 a variable number of temporal bases in CLIP latent space via an expectation-maximization (EM)
37 procedure with MDL/BIC model selection and train a compact temporal basis encoder to predict
38 per-time-step responsibilities. Each attention block then mixes basis-specific low-rank directions

with learned magnitudes in a DoRA-style decomposition, optionally parameter-compressed with Householder reflections to preserve orthogonality at low cost Liu et al. [2024], Dong et al. [2024]. The effective rank thus grows linearly with the selected number of bases, allowing longer or more dynamic videos to receive more adaptation capacity, while storage and inference remain light. Our use of EM responsibilities is inspired by spatial-temporal compaction through mixture modeling in video inversion Li et al. [2023]; here, it serves as a teacher to amortize routing at inference.

To integrate AB-LoRA into Video-LLMs, we adopt a three-stage pipeline aligned with recent video-language pretraining practices. Stage-1 aligns visual features to the frozen LLM token space. Stage-2 pretrains the temporal basis encoder on WebVid-scale data using MDL-GMM responsibilities as a teacher Bain et al. [2021]. Stage-3 instruction-tunes only the AB-LoRA parameters while freezing both the foundation model and the basis encoder, thereby limiting trainable weights and stabilizing training Jin et al. [2024]. Our evaluation plan spans Video QA (MSR-VTT-QA, MSVD-QA, ActivityNet-QA), temporal grounding (Charades-STA, ActivityNet-Captions), static Doc/Chart-QA to verify language preservation, and a long-video split. Efficiency is assessed by trainable parameters, peak GPU memory, and wall-time; robustness is probed along video length, motion complexity, and domain shift. Baselines include LoRA-8, DoRA-64, and a SimDA-like $K=1$ variant Hu et al. [2021], Liu et al. [2024], Xing et al. [2023].

AB-LoRA is complementary to representation-level advances such as query-aware keyframe selection, chained localization-answering, and boundary-aware training Yang et al. [2022], Yu et al. [2023], Huang et al. [2023], Xu et al. [2024]. It targets the adaptation mechanism itself, enabling capacity to scale with temporal complexity without altering inference-time latency or substantially increasing storage. This design resonates with theoretical insights on the role of rank in expressive power Zeng and Lee [2023] and with orthogonal parameterizations for efficient adaptation Dong et al. [2024].

We validate feasibility through a minimal synthetic sanity check that exercises the entire pipeline. As expected given the tiny backbone and training budget, the quick test does not validate the central hypothesis but surfaces calibration points for the MDL teacher and confirms end-to-end trainability. The full benchmark suite and ablations are prepared for execution.

- **Adaptive mixture-of-bases PEFT:** A video-adaptive mixture-of-bases PEFT mechanism whose effective rank scales with temporal complexity while storage remains bounded.
- **Temporal basis encoder:** A temporal basis encoder trained to imitate MDL-GMM responsibilities computed on CLIP features, amortizing per-video routing Bain et al. [2021], Li et al. [2023].
- **DoRA with Householder compression:** A DoRA-style direction-magnitude parameterization combined with Householder-inspired compression for a strong capacity-efficiency trade-off Liu et al. [2024], Dong et al. [2024].
- **Training and evaluation blueprint:** A practical three-stage training recipe, fairness protocols, robustness grids, and PyTorch skeletons enabling reproducible evaluation across Video QA, temporal grounding, and long-video reasoning.
- **Quick end-to-end validation:** A minimal end-to-end quick test establishing feasibility and informing subsequent calibration.

Looking ahead, we foresee sharing temporal bases across users for personalization and applying MDL-driven adaptive bases to video generation and editing, where temporally coherent latent structures are equally beneficial Guo et al. [2023], Xing et al. [2023], Li et al. [2023].

2 Related Work

Parameter-efficient fine-tuning. LoRA injects low-rank updates into frozen weights, drastically reducing trainable parameters and memory while maintaining competitive quality Hu et al. [2021]. However, its fixed rank limits expressivity under diverse inputs. Theory formally connects expressive power to rank, clarifying why fixed small ranks can be limiting Zeng and Lee [2023]. DoRA decomposes weight updates into a unit-norm direction and a scalar magnitude, improving capacity and stability without inference overhead Liu et al. [2024]. Flora interprets LoRA as gradient compression and proposes resampling to achieve high-rank updates with sublinear optimizer state memory Hao et al. [2024]. Our approach differs conceptually and operationally: instead of a single

shared low-rank subspace, we allocate multiple basis-specific low-rank directions and mix them per time step via video-adaptive responsibilities, making the effective rank grow with temporal complexity while the stored parameters remain modest.

Temporal modeling in video-language. FrozenBiLM adapts frozen bidirectional language models to video QA via masked language modeling with lightweight multimodal connectors, achieving strong zero-shot performance Yang et al. [2022]. SeViLA chains a localizer and an answerer within a single image-language model to obtain language-aware keyframes and alleviate annotation demands Yu et al. [2023]. VTimeLLM introduces a boundary-aware curriculum to improve temporal boundary understanding Huang et al. [2023]. Slot-VLM proposes object- and event-wise slots via a SlowFast design that yields semantically decomposed video tokens Xu et al. [2024]. These methods focus on representation and inference, whereas AB-LoRA focuses on the adaptation mechanism, and can be layered atop such systems to better allocate capacity to temporally complex inputs.

Video pretraining and tokenization. Frozen-in-Time demonstrates joint image-video encoders and introduces WebVid, enabling scalable video-text pretraining Bain et al. [2021]. Video-LaVIT decouples visual and motion tokenization for unified pretraining across images, videos, and text Jin et al. [2024]. We leverage CLIP features and WebVid-style data for basis pretraining and adopt a Stage-1 alignment to connect vision to the frozen LLM.

Adaptive decomposition and EM. STEM inversion models dense video features via a compact basis set learned with EM, improving temporal consistency for video editing Li et al. [2023]. We adopt the insight that mixture modeling captures temporal structure and use MDL-selected GMM responsibilities as a teacher to train a fast temporal basis encoder. Unlike STEM, our goal is parameter-efficient adaptation for reasoning rather than inversion or editing.

Orthogonal parameterizations. Householder transformations provide compact orthogonal factors and layer-wise scalable ranks for efficient adaptation of ViTs Dong et al. [2024]. We employ Householder-inspired compression for basis-specific directions to reduce parameters while approximately preserving orthogonality, complementing DoRA’s decomposition Liu et al. [2024].

Temporal adapters in diffusion. AnimatedDiff and SimDA validate lightweight temporal modules for adapting image diffusion models to video, emphasizing temporal consistency under PEFT Guo et al. [2023], Xing et al. [2023]. While these works address generation rather than reasoning, they motivate PEFT designs that respect temporal structure. Our $K=1$ ablation recovers a SimDA-like shared adapter, isolating the incremental value of adaptive bases.

3 Background

Problem setting and notation. We consider a Video-LLM composed of a frozen language backbone and a frozen vision encoder (e.g., CLIP ViT/B-16) that produces per-frame latent features. Given a video with T frames, let F_t denote the latent for frame t . A lightweight projection and positional fusion module converts $\{F_t\}_{t=1}^T$ into visual tokens and fuses them with text tokens in a causal transformer for instruction following or question answering.

PEFT preliminaries. LoRA adapts a pre-trained weight matrix W by learning a low-rank update $\Delta W = AB^\top$ with inner dimension r ; at inference $W + \Delta W$ is used Hu et al. [2021]. The rank r is fixed per module, which can be restrictive under varying temporal complexity. DoRA decomposes updates into a unit-norm direction and a scalar magnitude, enhancing learning capacity and stability while retaining PEFT efficiency Liu et al. [2024]. The expressive power of low-rank updates is fundamentally tied to available rank Zeng and Lee [2023].

MDL-guided mixture modeling. To capture temporal heterogeneity within a video, we consider a mixture model over per-frame latents. For each video, we fit diagonal-covariance Gaussian mixture models (GMMs) in a PCA-reduced latent space and select the number of components K via MDL/BIC. The posterior responsibilities $\gamma_{t,k}$ provide soft assignments indicating which temporal basis is active at time t . This idea is related to STEM’s EM-based compaction for video inversion Li et al. [2023], but here the responsibilities serve as a teacher signal for amortized routing.

Householder parameterization. Orthogonal or unitary matrices can be represented compactly via sequences of Householder reflections, enabling parameter-efficient approximations of unitary factors

and flexible ranks Dong et al. [2024]. We adopt Householder-inspired compression to reduce the parameter count for basis-specific directions while preserving desirable orthogonality.

Evaluation axes. We focus on: (1) video QA accuracy; (2) temporal grounding quality measured by mean IoU and recall at $\text{IoU} \geq 0.5$; (3) captioning metrics such as BLEU and CIDEr where applicable; (4) temporal consistency measured as average cosine similarity of adjacent-frame features; and (5) efficiency metrics including trainable parameters, peak GPU memory, and wall-time per epoch. These axes directly target known limitations in temporal order understanding and boundary precision Yang et al. [2022], Yu et al. [2023], Huang et al. [2023].

4 Method

4.1 Adaptive Mixture-of-Bases Overview

AB-LoRA replaces a single static low-rank adapter in each targeted projection with a mixture of K basis-specific low-rank directions whose combination is modulated over time by per-frame responsibilities. Responsibilities are predicted by a temporal basis encoder trained to imitate MDL-selected GMM responsibilities computed over CLIP latents. Each basis direction uses a DoRA-style decomposition into direction and magnitude, with optional Householder-inspired compression.

4.2 Mixture-of-Bases Adapter Construction

Consider an attention block with hidden size D applied to T visual tokens $X \in \mathbb{R}^{T \times D}$. We instantiate K_{max} basis slots indexed by k . Each basis k carries a rank- r direction update $\Delta V_k \in \mathbb{R}^{D \times D}$ and a scalar magnitude m_k . For a given video, the temporal basis encoder outputs logits over K_{max} per time step, converted into responsibilities $\gamma_{t,k}$ by a temperature-controlled softmax. If the MDL-selected number of bases K_{sel} is less than K_{max} , responsibilities for $k > K_{\text{sel}}$ are zeroed. The time-varying update for step t is

$$\Delta W_t = \sum_{k=1}^{K_{\text{max}}} \gamma_{t,k} m_k \Delta V_k.$$

The residual-style application is

$$Y_t = X_t(I + \Delta W_t^\top),$$

with effective rank per video $r \cdot K_{\text{sel}}$ while storage scales as $r \cdot K_{\text{max}}$ with K_{max} bounded (e.g., ≤ 256). Thus, capacity grows with temporal complexity while memory remains bounded.

4.3 Temporal Basis Encoder

The encoder is a compact 1D CNN over the sequence $\{F_t\}$ with two convolutional layers, a 1×1 projection to K_{max} , and an exponential moving average smoother to stabilize logits. A learned temperature controls responsibility sharpness. The encoder is trained as a student to match MDL-GMM responsibilities from the teacher by minimizing a per-video KL divergence.

4.4 Teacher Responsibilities via EM with MDL/BIC

For each training video in basis pretraining, we: (a) PCA-reduce frame latents to a compact subspace; (b) fit diagonal-covariance GMMs for candidate K values; (c) select K by MDL/BIC; and (d) compute posterior responsibilities $\gamma_{t,k}$. The encoder is trained to minimize per-video KL divergence between its predicted distribution (restricted to K_{sel}) and the teacher, amortizing per-video EM at inference by replacing it with one forward pass.

4.5 DoRA Decomposition and Householder Compression

Each ΔV_k is parameterized by low-rank factors that are L2-normalized to represent a direction, and m_k is a learned scalar magnitude, following DoRA to improve learning capacity without inference overhead Liu et al. [2024]. We further apply Householder-inspired compression to parameterize orthogonal factors efficiently, reducing parameters while preserving the ability to represent rotations in feature space Dong et al. [2024].

184 4.6 Training Regime

185 Stage-1 Feature Alignment: with the language model frozen, train a visual projection adapter and
 186 positional fusion to align CLIP features to the LLM token space, following unified video-language
 187 pretraining practices Jin et al. [2024]. Stage-2 Temporal Basis Pretraining: freeze the LLM and
 188 adapters and train only the temporal basis encoder on WebVid-scale data using MDL-GMM teacher
 189 responsibilities, minimizing KL divergence per video Bain et al. [2021]. Stage-3 Instruction Tuning:
 190 freeze the basis encoder and train only the AB-LoRA parameters (basis-specific directions and
 191 magnitudes) in selected attention blocks, optimizing on curated instruction and QA data while
 192 keeping all other weights frozen.

193 4.7 Ablations and Diagnostics

194 We include ablations that: (i) disable MDL by fixing K ; (ii) remove DoRA to use pure low-rank bases;
 195 (iii) set $K = 1$ to emulate SimDA-like temporal sharing Xing et al. [2023]; (iv) toggle Householder
 196 compression; and (v) vary rank r . Diagnostics monitor responsibility calibration (ECE and KL to the
 197 teacher), the distribution of selected K versus motion entropy and video length, magnitude sparsity
 198 of m_k , and effective rank distributions.

199 4.8 Computational Profile

200 AB-LoRA adds a small temporal encoder forward pass and a per-step mixture of tiny low-rank
 201 matrices. Memory grows with $r \cdot K_{\max}$ but remains bounded and comparable to standard PEFT
 202 configurations, while effective rank scales with the selected K per video.

203 4.9 Algorithms

204 MDL-selected EM teacher for responsibilities Input: per-frame latents $\{F_t\}_{t=1}^T$, candidate set
 205 $\mathcal{K} = \{1, \dots, K_{\max}\}$ Compute PCA projection: $\tilde{F}_t \leftarrow PCA(F_t)$ $K \in \mathcal{K}$ Fit diagonal-covariance
 206 GMM with K components via EM on $\{\tilde{F}_t\}$ Record MDL/BIC score $MDL(K)$ Select $K_{\text{sel}} \leftarrow$
 207 $\arg \min_K MDL(K)$ Refit GMM with K_{sel} and compute posteriors $\gamma_{t,k}$ Output: responsibilities
 208 $\{\gamma_{t,k}\}$ and K_{sel}

209 AB-LoRA forward in one attention block Input: tokens $X \in \mathbb{R}^{T \times D}$, basis params $\{\Delta V_k, m_k\}_{k=1}^{K_{\max}}$,
 210 basis encoder E Compute logits: $\ell_{t,k} \leftarrow E(\{F_t\})$ Responsibilities: $\gamma_{t,k} \leftarrow \text{softmax}_k(\ell_{t,k}/\tau)$; zero
 211 out $k > K_{\text{sel}}$ $t = 1$ to T $\Delta W_t \leftarrow \sum_{k=1}^{K_{\max}} \gamma_{t,k} m_k \Delta V_k$ $Y_t \leftarrow X_t(I + \Delta W_t^\top)$ Output: Y

212 5 Experimental Setup

213 Base models and injection points. We instantiate AB-LoRA on a frozen Vicuna-7B language backbone
 214 with a CLIP ViT/B-16 vision encoder, using a lightweight visual projection and positional fusion
 215 module to produce visual tokens. AB-LoRA modules are injected into selected attention projection
 216 matrices. Baselines include LoRA-8, DoRA-64, and a SimDA-like variant implemented as $K = 1$
 217 with shared temporal adapters Hu et al. [2021], Liu et al. [2024], Xing et al. [2023].

218 Datasets and tasks. Planned benchmarks cover diverse temporal reasoning demands:

- 219 • Video QA: MSR-VTT-QA, MSVD-QA, ActivityNet-QA; official splits when available.
- 220 • Temporal grounding: Charades-STA and ActivityNet-Captions, prompting the model to
- 221 output time spans.
- 222 • Static language preservation: DocVQA and ChartQA to verify no regression on static inputs.
- 223 • Long-video reasoning: a long-video split of EgoSchema with 2-3 minute videos.

224 Sampling and features. Videos are decoded with uniform frame sampling. AB-LoRA targets 64
 225 frames; baselines operate at 16 frames under equal-memory settings. A matched-frames control
 226 (16-frame AB-LoRA) disentangles method gains from increased context. CLIP features are computed
 227 per frame and normalized. The temporal basis encoder includes an EMA downsampler.

228 Training protocol. Stage-1 (feature alignment) runs for one epoch with the LLM frozen. Stage-2
 229 (basis pretraining) runs for two epochs using MDL-GMM teacher responsibilities computed in
 230 PCA-reduced CLIP latent space, minimizing KL divergence per video Bain et al. [2021]. Stage-3
 231 (instruction tuning) runs for two epochs on a mixture of Video-ChatGPT, VideoInstruct100k, and
 232 benchmark QA data, with the basis encoder frozen and only AB-LoRA parameters and the visual
 233 adapter trainable. We use AdamW, cosine learning-rate decay, gradient clipping, mixed precision, and
 234 distributed data parallelism. Memory parity across methods is enforced via gradient accumulation
 235 and frame-count control; we report peak memory and wall-time per epoch.

236 Metrics. We evaluate QA accuracy; BLEU and CIDEr for captioning where applicable; temporal
 237 grounding using mean IoU and recall at $\text{IoU} \geq 0.5$; temporal consistency via average adjacent-frame
 238 feature cosine; and efficiency via trainable parameters, peak GPU memory, and wall-time per epoch.
 239 Robustness is measured across bins of video length (10-180 s), motion complexity quartiles defined
 240 by optical-flow entropy, and domain shifts (e.g., recipes to sports).

241 Ablations and diagnostics. Ablations include MDL on/off (fixed K), DoRA on/off, $K = 1$ (SimDA-
 242 equivalent), Householder compression on/off, and rank $r \in \{2, 4, 8\}$. Diagnostics include responsi-
 243 bility calibration (ECE and KL to the MDL teacher), K distributions versus flow entropy and video
 244 length, magnitude sparsity of m_k , effective rank distributions, and static-task accuracy to confirm no
 245 language regression Xu et al. [2024], Huang et al. [2023].

246 Minimal sanity check actually executed. To validate end-to-end wiring before large-scale training,
 247 we conducted a tiny synthetic quick test using the provided skeletons. Setup: a mock CLIP-style
 248 feature extractor (tiny CNN; hidden dimension 64), AB-LoRA with rank $r = 2$ and $K_{\text{max}} = 8$,
 249 and a synthetic shapes dataset of 12 training and 8 evaluation videos. Videos contained colored
 250 objects moving with piecewise-constant velocities; QA was a 4-way classification of the number of
 251 motion segments; temporal grounding estimated the time of the first change. Stage-2 basis pretraining
 252 ran for 6 steps to match MDL-GMM responsibilities; Stage-3 adapter training ran for 12 steps
 253 on the synthetic QA/grounding supervision. Responsibilities were computed via MDL-GMM on
 254 PCA-reduced features at inference. Recorded metrics: QA accuracy, temporal grounding mean IoU,
 255 temporal consistency (adjacent-frame feature cosine), mean selected K , and mean flow entropy. No
 256 baselines or standard datasets were executed in this quick test, and no confidence intervals were
 257 computed due to the extremely small sample size.

258 6 Results

259 We report only the results executed and logged in the minimal synthetic sanity check. Stage-2 basis
 260 pretraining completed in 6 steps with a final training loss approximately 0.0000, indicating near-
 261 perfect imitation of the MDL-GMM teacher on the tiny dataset. Stage-3 adapter training completed
 262 in 12 steps with a final loss of 3.1638. On 8 synthetic evaluation videos, we obtained: QA accuracy
 263 of 0.375 (4-way classification), temporal grounding mean IoU of 0.11666666664472222, temporal
 264 consistency of 0.9998192435791415 (adjacent-frame feature cosine), mean selected K of 1.0, and
 265 mean flow entropy of 0.30935862273788595.

266 Interpretation. The quick test demonstrates that the end-to-end AB-LoRA pipeline executes: MDL-
 267 based teacher responsibilities are computed, the temporal basis encoder can learn to predict them,
 268 and the mixture-of-bases adapter is trainable under supervision. Capacity scaling was not observed:
 269 the MDL criterion selected a single basis on average ($K_{\text{mean}} = 1.0$). This collapse is consistent
 270 with the tiny feature capacity of the mock backbone, the extremely small training budget, and
 271 limited motion separability in the synthetic features. Task performance is weak by design and is not
 272 comparable to the planned benchmarks. As no baselines were run in this sanity check, no comparative
 273 claims can be made.

274 Fairness, hyperparameters, and robustness. The quick run did not execute memory-matched baselines,
 275 the robustness grid, or long-video evaluations; accordingly, fairness controls (peak memory, wall-
 276 time parity) and confidence intervals are deferred to the full experiments. The run used $r = 2$,
 277 $K_{\text{max}} = 8$, and responsibilities computed via MDL-GMM on PCA-reduced features. Temporal
 278 consistency was measured as average adjacent-frame feature cosine without optical-flow warping.

279 Limitations. The minimal setup-tiny backbone, minimal steps, and very small datasets-cannot validate
 280 the central hypothesis that adaptive effective rank improves temporal grounding and consistency. The

281 absence of baselines and standard datasets further limits interpretation. These results serve only as
282 pipeline validation and MDL calibration.

283 7 Conclusion

284 We presented Adaptive Basis LoRA (AB-LoRA), a parameter-efficient fine-tuning method for
285 Video-LLMs that allocates adaptation capacity in proportion to the temporal complexity of the input.
286 AB-LoRA replaces a single static low-rank update with a mixture of basis-specific low-rank directions
287 whose combination over time is predicted by a temporal basis encoder trained to imitate MDL-GMM
288 responsibilities computed in CLIP latent space Bain et al. [2021], Li et al. [2023]. The design
289 combines DoRA’s direction-magnitude decomposition with Householder-inspired compression to
290 strengthen the capacity-efficiency trade-off while maintaining the inference efficiency characteristic
291 of PEFT Liu et al. [2024], Dong et al. [2024]. We detailed a three-stage training recipe, evaluation
292 protocols, fairness controls, robustness grids, and ablations, and provided PyTorch skeletons enabling
293 reproducibility.

294 A minimal synthetic sanity check demonstrated end-to-end feasibility but did not test the core
295 hypothesis. Immediate next steps are to execute the full experimental plan: pretrain the basis encoder
296 with MDL-GMM teachers on WebVid-scale data, instruction-tune with the encoder frozen, and
297 evaluate against LoRA-8, DoRA-64, and SimDA-like baselines under memory-matched settings
298 across Video QA, temporal grounding datasets, static Doc/Chart-QA, and long-video scenarios Hu
299 et al. [2021], Liu et al. [2024], Xing et al. [2023]. Ablations will isolate the effects of MDL-based
300 K selection, DoRA, Householder compression, and rank r ; diagnostics will verify that selected K
301 correlates with motion entropy and video length. If confirmed, AB-LoRA would provide a practical,
302 plug-and-play route to temporally consistent, long-video reasoning within tight memory budgets and
303 without additional inference latency. Longer term, we see opportunities to share bases across users
304 for personalization and to apply MDL-driven adaptive bases to video generation and editing where
305 temporally coherent latent structures are similarly beneficial Guo et al. [2023], Li et al. [2023].

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[width=0.7] images/confusion_matrix_test.pdf

Figure 1: Confusion matrix for the quick-test 4-way QA classification in the minimal setup (filename: confusion_matrix_test.pdf).

0.48 [width=] images/training_loss_ab_lora_pair1.pdf

Figure 2: AB-LoRA training loss curve showing convergence over 12 steps (filename: training_loss_ab_lora_pair1.pdf).

0.48 [width=] images/training_loss_baselines_pair2.pdf

Figure 3: Baseline training-loss artifact produced by the logging pipeline (filename: training_loss_baselines_pair2.pdf).